AWS SQL Server — Licensing & Pricing Guide
Amazon RDS & Amazon EC2 (SQL Server Workloads)

Compute Models • Licensing • Scaling • Cost Optimization • HA/DR

1. What is AWS RDS for SQL Server?

Amazon RDS for SQL Server is a **fully managed PaaS database service** running Microsoft SQL Server.

Key Characteristics

- AWS manages:
    - Hardware provisioning
    - OS patching
    - SQL Server installation & patching
    - Automated backups & snapshots
    - High availability (Multi-AZ)
- You manage:
    - Database schema & indexing
    - Query performance tuning
    - Security (IAM, encryption policies)

SLA

- Up to **99.99% availability** (Multi-AZ deployments)

Deployment Types

- Single-AZ (dev/test)
- Multi-AZ (production, synchronous standby)

2. What is AWS EC2 for SQL Server?

Amazon EC2 for SQL Server is an **IaaS model** where you run SQL Server on virtual machines.

Key Characteristics

- Full control over:
    - OS, SQL Server version, patches
    - Storage layout, tempdb tuning
    - High availability architecture
- You are responsible for:
    - Backups
    - HA/DR setup
    - OS & DB patching

Best Fit

- Legacy apps requiring OS-level control
- Custom SQL Server configurations
- Third-party tools requiring admin access

3. Licensing Models (Critical Difference)

RDS Licensing Options

License Included (LI)

- SQL Server license bundled into hourly rate
- Editions available:
    - Express (free)
    - Web
    - Standard
    - Enterprise

Bring Your Own License (BYOL)

- Use existing SQL Server licenses with Software Assurance
- Only supported on:
    - RDS Custom
    - EC2

EC2 Licensing Options

License Included (LI)

- Pay hourly (compute + SQL license)

BYOL (Highly Flexible)

- Use:
    - SQL Server Standard / Enterprise
    - Core-based licensing
- Requires **License Mobility (Software Assurance)**

4. Instance-Based Compute Model (RDS Equivalent to vCore)

AWS does NOT use DTU/vCore abstraction.

Instead:

- You choose **instance types** (CPU + RAM predefined)

Common Instance Families

Family	Best For
db.t3 / db.t4g	Burstable workloads, dev/test
db.m5 / db.m6i	General-purpose OLTP
db.r5 / db.r6i	Memory-intensive workloads
db.x1 / db.x2idn	Enterprise-scale, large DBs

Key Differences vs Azure

- No abstract metric like DTU
- Full visibility of:
    - vCPU count
    - RAM
    - Network throughput

5. Storage Model

RDS Storage Types

Type	Description	Best Use
General Purpose (gp3)	Balanced SSD	Most workloads
Provisioned IOPS (io1/io2)	High-performance SSD	Critical OLTP
Magnetic (legacy)	Low cost	Rare use

Limits

- Storage: up to **64 TB**
- IOPS: up to **256,000 (io2 Block Express)**

EC2 Storage (More Flexible)

- Amazon EBS (gp3, io2)
- Instance store (ephemeral, ultra-fast)
- Full control over:
  - RAID setups
  - Filegroup distribution

6. High Availability & Disaster Recovery

RDS HA Options

Multi-AZ (Primary + Standby)

- Synchronous replication
- Automatic failover
- No manual intervention

Read Replicas

- Not supported for SQL Server (important limitation)

EC2 HA Options

- SQL Server Always On Availability Groups
- Failover Cluster Instances (FCI)
- Log shipping / replication

Key Difference

Feature	RDS	EC2
HA Setup	Automatic	Manual
Failover	Managed	You manage
Flexibility	Limited	Full control

7. Backup & Recovery

RDS

- Automated backups (up to 35 days)

- Point-in-time recovery
- Snapshots (manual)

EC2

- You design:
  - Backup strategy (native SQL backups)
  - Snapshot automation
  - Retention policies

8. Scaling Model

RDS Scaling

Vertical Scaling (Primary method)

- Change instance type
- Downtime: minimal (few minutes)

Storage Scaling

- Can scale without downtime

EC2 Scaling

- Vertical scaling (instance resize)
- Horizontal scaling (AG replicas, sharding)

9. Pricing Model (2025–2026 AWS Structure)

RDS Pricing Components

- Instance hours (vCPU + RAM)
- Storage (GB/month)
- IOPS (if provisioned)
- Backup storage beyond free tier

Example (approximate baseline)

- db.m6i.large (2 vCPU, 8 GB RAM)
  - ~$0.20–$0.40/hr (license included varies by edition)

EC2 Pricing Components

- EC2 compute cost
- SQL Server license (if LI)
- EBS storage
- Data transfer

10. Cost Optimization Strategies

RDS

- Reserved Instances (1-year / 3-year)
  - Save up to **60%**
- Stop/start (non-production)

- Right-size instances

EC2

- BYOL = biggest savings lever
- Savings Plans / Reserved Instances
- Use Spot (non-critical workloads)

11. RDS vs EC2 — Head-to-Head Comparison

Criteria	RDS (PaaS)	EC2 (IaaS)
Management	AWS-managed	Self-managed
OS Access	No	Full
Patching	Automatic	Manual
HA Setup	Built-in	Manual
Scaling	Easy	Complex
Customization	Limited	Full
Licensing	LI + limited BYOL	Full BYOL flexibility
Best For	Standard workloads	Enterprise / custom setups

12. When to Choose What (Decision Framework)

Choose RDS for SQL Server if:

- You want **minimal operational overhead**
- Standard OLTP workloads
- No need for OS-level access
- Fast deployment required

Choose EC2 for SQL Server if:

- You need:
  - SQL Server Agent customization
  - SSIS / SSRS / SSAS full support
- You require:
  - Always On advanced configurations
  - Third-party integrations
- You want **maximum licensing control (BYOL)**

13. Key Differences vs Azure SQL Database

Feature	Azure SQL DB	AWS RDS SQL Server
Model	PaaS (fully abstracted)	PaaS (instance-based)
Scaling	vCore / DTU	Instance type

Feature	Azure SQL DB	AWS RDS SQL Server
Serverless	Yes	Limited (no true serverless SQL Server)
Max Storage	100 TB (Hyperscale)	64 TB
Read Replicas	Yes	No (SQL Server limitation in RDS)

14. Practical Architecture Patterns

RDS Pattern

- Web/App → RDS Multi-AZ
- Backup → S3 snapshots
- Monitoring → CloudWatch

EC2 Pattern

- App → SQL Server Always On AG
- Shared storage (EBS)
- Backup → S3 via scripts

15. Summary (Executive View)

- **RDS = Managed convenience**
- **EC2 = Maximum control**
- **Biggest cost lever = licensing (BYOL vs LI)**
- **Biggest architectural decision = control vs simplicity**

AWS SQL Server – Decision Tree for Architects

Choose the Right Deployment, Licensing & Pricing Strategy

START
Where will your SQL Server workload run?

PREFER MANAGED SERVICE
(Minimal operational overhead)
Consider **AMAZON RDS** for SQL Server

Do you need OS-level access, custom configurations, or third-party software that requires admin access?

NO

Choose RDS for SQL Server (Minaged PaaS)
- AWS manages OS, patching, backups, HA, monitoring
- Multi-AZ for high availability
- Up to 64 TB storage
- No OS access (by design)
- No read replicas support for SQL Server

YES

Choose RDS Custom (Managed OS access)
- Full OS access on RDS platform
- Bring Your Own License (BYOL) only
- You manage OS & SQL patches
- Same automation (backups, HA infrastructure) as RDS
- Ideal for custom configs or software requirements

NEED FULL CONTROL
(Maximum flexibility)
Consider **AMAZON EC2** for SQL Server

Do you need maximum licensing flexibility (BYOL) or advanced SQL Server features/configurations?

NO

EC2 with License Included (Quick start option)
- Pay hourly (includes SQL license)
- AWS manages NOTHING (you manage OS, SQL, backups, HA)
- Good for simple deployments without existing licenses
- Higher ongoing cost vs BYOL

YES

EC2 with Bring Your Own License (BYOL)
- Use existing SQL Server licenses with Software Assurance
- Significant cost savings
- Full control over OS, SQL, patching, HA/DR, tuning
- Best for enterprise and custom requirements

RDS – Choose the Right Instance Type

Instance Family	Best For
db.t3 / db.t4g	Burstable workloads, dev/test
db.m5 / db.m6i	General purpose, OLTP
db.r5 / db.r6i	Memory intensive workloads
db.x1 / db.x2idn	Enterprise scale, large DBs

RDS Storage Options (Up to 64 TB)
- **gp3** – General Purpose SSD, Balanced performance and cost
- **io1 / io2** – Provisioned IOPS SSD, High performance OLTP workloads
- **Magnetic** (HDD, legacy) – Low throughput, rarely used

EC2 – Choose the Right Compute & Storage

Compute (Instance Families)	Storage (Highly Flexible)
M / R / X / Z General purpose to memory optimized	**EBS gp3** General Purpose SSD
C / C7i Compute optimized	**EBS io2 / io2 Block Express** High performance SSD (up to 256K IOPS)
I / D I/O and storage optimized	**Instance Store (Ephemeral NVMe SSD)** Ultra-fast temporary storage
P / G Accelerated / GPU (rare for SQL Server)	**RAID / SQL Filegroups** Full control over layout

High Availability (Choose Your Option)

RDS for SQL Server	RDS Custom / EC2
Multi-AZ (synchronous standby)	SQL Server Always On AG
Automatic failover	Failover Cluster Instances (FCI)
Up to 99.99% availability	Log shipping / Replication
No manual intervention	You manage failover

Backup & Recovery

RDS for SQL Server	EC2 for SQL Server
Automated backups (1–35 days)	Design your backup strategy
Point-in-time recovery	Native SQL backups
Manual snapshots	EBS snapshots to S3
Backup storage managed by AWS	You manage retention

Licensing Options Summary

RDS for SQL Server
- License Included (LI) (Express, Web, Standard, Enterprise)
- BYOL only on RDS Custom

EC2 for SQL Server
- License Included (LI)
- BYOL (requires Software Assurance / License Mobility)

Cost Optimization

RDS
- 1-yr / 3-yr Reserved Instances (up to ~60% savings)
- Right-size instances
- Stop/start non-production instances
- Use gp3 storage (cost-effective)

EC2
- BYOL for significant savings
- Reserved Instances / Savings Plans
- Use right-sized instances
- Use Spot Instances for non-critical workloads

Final Recommendation Guide

Choose RDS for SQL Server if:
You want managed service, quick deployment, minimal operational overhead.

Choose RDS Custom if:
You need OS access but want managed infrastructure and automation.

Choose EC2 for SQL Server if:
You need full control, advanced configurations, BYOL flexibility, or custom HA/DR setups.

KEY TAKEAWAY: RDS = Managed Simplicity | RDS Custom = Managed with Control | EC2 = Maximum Control & Flexibility
Choose based on your workload needs, operational model, and licensing strategy.